I have completed my PhD in astronomy and astrophysics at the University of Geneva, where I designed and built NIGHT, a near-infrared, high-resolution spectrograph for helium detection in exoplanet atmospheres. My research combines optical engineering, spectroscopy, and observational astronomy to study exoplanetary atmospheric escape. Currently, I continue to be project lead on NIGHT as a postdoctoral researcher at the University of Geneva.

Personal Details

Rue John-Grasset 8
1205 Genève
Switzerland

Nationality: Dutch (NL)
Swiss Permit: B

M:
E(w):
E(p):

website: **casperfarret.github.io**

Skills & Interests

Astronomy & Astrophysics
Instrumentation
Engineering
Optics/lithography
Informatics
Public speaking/outreach
Teaching
Leadership
Graphic design

Languages

Dutch (native)
English (fluent)
German (proficient)
French (basic)

Programming & Software

Python
MATLAB
LaTeX
Zemax
COMSOL
Adobe CC

Casper (C.C.) Farret Jentink

Curriculum Vitae

Academic Qualifications

- Département d 'Astronomie - Université de Genève — 2021 — 2025
  PhD in Astronomy & Astrophysics
  advised by Prof. F. Pepe & Prof. C. Lovis
  - Led an international team of scientists and engineers.
  - Designed and built NIGHT: a high-resolution, near-infrared spectrograph.
  - Secured 135k CHF in funding for development of new instrumentation.
  - Secured 100k+ CHF of in-kind hardware contributions to NIGHT.
  - Collaborated with industrial partners to develop novel instrumentation for fiber-fed, high-resolution spectroscopy.
  - Secured 60 nights of telescope time for NIGHT at Observatoire de Haute Provence.
  - Supervised and mentored students and taught courses on astrophysics and instrumentation to master's students.
- Kapteyn Astronomical Institute - University of Groningen — 2018 — 2021
  MSc in Astronomy
  - Thesis on development of ABORAS: a polarimetric sun-as-a-star telescope for HARPS3.
    Supervised by: Prof. F. Snik & Dr. W. Jellema
- Kapteyn Astronomical Institute - University of Groningen — 2015 — 2018
  BSc in Astronomy

Professional Experience

- **Postdoctoral researcher** — 2025 — present
  Département d 'Astronomie - Université de Genève
  - Continuation as NIGHT project lead.
- Board member — 10.2019 — 08.2021
  JWG – Dutch youth association for astronomy
- Blaauw Observatory assistant — 02.2017 — 08.2021
- Local Policy Coordinator on education & culture — 04.2021 — 08.2021
  VOLT Netherlands
- Research intern on micromachined dielectrics for SPICA/SAFARI — 02.2020 — 07.2020
  SRON / TU-Delft
- Member of the advisory board — 09.2019 — 09.2020
- Chairman of the board (Budget: ~20k EUR, >250 members) — 09.2018 — 09.2019
- Vice-Chairman & Commissioner of Internal Affairs — 09.2017 — 09.2018
  Sirius A – the Study Association of Astronomy

Awards, Distinctions & Grants

- NCCR PlanetS Open Call Grant (135k CHF) — 2023
- Alpbach Summer School award for the EXPOSURE mission — 2023
- Beta Business Days Tech Start-up Award — 2021
- Kapteyn Science Day Award — 2021

Research Activity

- Project lead NIGHT: Near-Infrared Gatherer of Helium Transits[1] (Budget: ~900k CHF) — 2021 — present
- Project lead ABORAS: a polarimetric Sun-as-a-star telescope connected to HARPS3[2] (Budget: ~100k EUR) — 2021
- Simulation & Evaluation of an Effective Dielectric Material for a Resonant Immersed Waveplate[3] — 2020
- Constructing the backend of the Kapteyn Interferometer for Short-baseline Solar observations — 2018

Teaching & Mentorship

- Teaching assistant for Observational Techniques — 2022 — 2025
  Prof. Francesco Pepe
  University of Geneva, Geneva, Switzerland
- Teaching assistant for Observations, Data Acq. & Analysis — 2023
  Prof. Laurent Eyer
  University of Geneva, Geneva, Switzerland
- Supervision of master's student A. Besse — 2023
  University of Geneva, Geneva, Switzerland
- DOTLiveplanetarium operator — 01.2018 — 08.2021
  University of Groningen, Groningen, Netherlands
- Kapteyn Mobile Planetarium operator — 04.2017 — 08.2021
  University of Groningen, Groningen, Netherlands
- Teaching assistant for the Evolving Universe course — 10.2020 — 01.2021
  University of Groningen, Groningen, Netherlands
- First year mentor — 09.2019 — 03.2020
  University of Groningen, Groningen, Netherlands
- Astrophysics teacher — 08.2018 — 11.2018
  Boswell Beta, Utrecht, Netherlands

Selected attended Conferences, Workshops & Events

SPIE Astronomical Telescopes + Instrumentation	Yokohama, Japan	June 2024
Observing run at the EULER telescope	La Silla, Chile	September 2023
Spectral Fidelity conference	Florence, Italy	September 2023
Alpbach Summer School	Alpbach, Austria	Summer 2023
Extreme Precision Radial Velocity V	Santa Barbara, CA	April 2022

Selected Invited Talks, Media & Outreach

Documentary for Dutch Public Broadcasting	VPRO/Tegenlicht	September 2025
Press release on NIGHT first light	University of Geneva	September 2025
Interview for Dutch national newspaper	NRC	April 2025
Talk at NCCR PlanetS General Assembly	Neuchatel, Switzerland	April 2025
Talk at Scientifika	Stockholm, Sweden	September 2024
Talk at NCCR PlanetS General Assembly	Engelberg, Switzerland	April 2024

References are available on request.

Publications

First-Author Publications

Farret Jentink, C., Pepe, F., Lovis, C., Schwab, C., Wildi, F., and Clawson, A. (2025). High-resolution, high-efficiency narrowband spectroscopy with an s-p-phased holographic grating in double pass. *Astronomy & Astrophysics*, 698, A249.

Farret Jentink, C., Pepe, F., Lovis, C., Bovay, S., Wildi, F., Chazelas, B., Sordet, M., Artigau, É., Doyon, R., Baron, F., Bourrier, V., Allart, R., and Cochard, F. (2024). The Near-Infrared Gatherer of Helium Transits (NIGHT). *Proceedings of SPIE*, 13096, 130967O.

Farret Jentink, C., Bourrier, V., Lovis, C., Allart, R., Chazelas, B., Lendl, M., Dumusque, X., and Pepe, F. (2024). NIGHT: A compact, near-infrared, high-resolution spectrograph to survey helium in exoplanet systems. *Monthly Notices of the Royal Astronomical Society*, 527(3), 4467–4482.

Farret Jentink, C., Mortier, A., Snik, F., Dorval, P., Thompson, S. J., Navarro, R., and Naylor, T. (2022). ABORAS: polarimetric, 10 cm/s RV observations of the Sun as a star. *Proceedings of SPIE*, 12182, 1218231.

Theses

Farret Jentink, C. (2025). The Design and Development of NIGHT: A High-Resolution Near-Infrared Spectrograph for Exoplanetary Atmospheric Escape Studies via Helium Triplet Spectroscopy. *PhD Thesis*.

Farret Jentink, C. (2021). A Design Study of ABORAS: A Robotic, Polarimetric Solar Telescope for HARPS3. *Master's Thesis*.

Farret Jentink, C. (2020). Simulation and Evaluation of an Effective Dielectric Material for a Resonant Immersed Waveplate. *Internship Report*.

Farret Jentink, C. (2018). Constructing the Backend of the Kapteyn Interferometer for Short-Baseline Solar Observations (KISS). *Bachelor's Thesis*.

Co-Author Publications

Seidel, J. V., Prinoth, B., Pino, L., dos Santos, L. A., Chakraborty, H., Parmentier, V., Sedaghati, E., Wardenier, J. P., **Farret Jentink, C.**, Zapatero Osorio, M. R., Allart, R., Ehrenreich, D., Lendl, M., Roccetti, G., Damasceno, Y., Bourrier, V., Lillo-Box, J., Hoeijmakers, H. J., Pallé, E., Santos, N., Suárez Mascareño, A., Sousa, S. G., Tabernero, H. M., and Pepe, F. A. (2025). Vertical structure of an exoplanet's atmospheric jet stream. *Nature*, 639(8056), 902–908.

Frensch, Y. G. C., Bouchy, F., Lo Curto, G., Ulmer-Moll, S., Sousa, S. G., Santos, N. C., Stassun, K. G., Watkins, C. N., Chakraborty, H., Barkaoui, K., Battley, M., Ceva, W., Collins, K. A., Daylan, T., Evans, P., Faria, J. P., **Farret Jentink, C.**, et al. (2025). Three hot Jupiters transiting K-dwarfs with significant heavy element masses. *Astronomy & Astrophysics*, 700, A118.

Fontanet, E., Udry, S., Ségransan, D., Figueira, P., Acevedo Barroso, J. A., Akinsanmi, B., Attia, M., Battley, M., Bhatnagar, S., Bugatti, M., Carteret, Y., Chakraborty, H., Deline, A., **Farret Jentink, C.**, et al. (2025). CORALIE radial-velocity search for companions around evolved stars (CASCADES): IV. New planetary systems around HD 87816, HD 94890, and HD 102888 and an update on HD 121056. *Astronomy & Astrophysics*, 699, A38.

Hagelberg, J., Restori, N., Ségransan, D., Guyon, O., Wildi, F., Chazelas, B., Baranec, C., Deo, V., Genolet, L., Sordet, M., Alesina, F., Algelly, M., Blind, N., Bonfils, X., Bouchy, F., Bovay, S., Buchschacher, N., Cantero, C., Carvalho, A., Ceva, W., Conod, U., Crausaz, A., Dezempte, L., **Farret Jentink, C.**, et al. (2024). On-sky performance of KalAO the swift adaptive optics imager on the 1.2m Euler Swiss telescope in La Silla, Chile. *Proceedings of SPIE*, 13097, 130970C.

Dorval, P., Politopoulos, I., Nicholson, B., ter Horst, R., **Farret Jentink, C.**, Klein, B., Kragt, J., Mortier, A., Navarro, R., Naylor, T., Peterzon, J. R., Piskunov, N., Rescigno, F., and Thompson, S. (2024). The polarimetric performance of HARPS3 and its impact on the Terra Hunting Experiment scheduling. *Proceedings of SPIE*, 13096, 130968C.

Jellema, W., Ridder, M., Eggens, M., **Farret Jentink, C.**, Formsma, J., Alfaro-Gomez, M., and Gutiérrez-Tadeo, G. (2022). A Broadband Resonant Half-Waveplate for Compact Far-Infrared Grating Spectrometers with Continuous Wavelength Band Division. *Proceedings of the 2022 47th International Conference on Infrared, Millimeter and Terahertz Waves (IRMMW-THz)*, 1–1.